Newman-Penrose Formalism Calculations

Stephani et al., Equation (20.54)

Metric

$$ds^2 = -\frac{A}{B}\mathrm{d}t \otimes \mathrm{d}t + \frac{C}{B}\mathrm{d}t \otimes \mathrm{d}\varphi + \frac{Be^{(2\,k)}}{A}\mathrm{d}u \otimes \mathrm{d}u + \frac{Be^{(2\,k)}}{A}\mathrm{d}v \otimes \mathrm{d}v + \frac{C}{B}\mathrm{d}\varphi \otimes \mathrm{d}t + \frac{G}{B}\mathrm{d}\varphi \otimes \mathrm{d}\varphi$$

Coordinates

$$(x^\mu) = (t, u, v, \varphi)$$

Metric Functions

$$A(u, v) = A$$

$$B(u, v) = B$$

$$C(u, v) = C$$

$$G(u, v) = G$$

$$k(u, v) = k$$

Null Tetrad

Covariant components

$$l = \left(\frac{\sqrt{2}\sqrt{A}}{2\sqrt{B}}\right) dt + \left(-\frac{\sqrt{2}C}{2\sqrt{AB}} + \frac{\sqrt{2}\sqrt{C^2 + AG}}{2\sqrt{AB}}\right) d\varphi$$

$$n = \left(\frac{\sqrt{2}\sqrt{A}}{2\sqrt{B}}\right) dt + \left(-\frac{\sqrt{2}C}{2\sqrt{AB}} - \frac{\sqrt{2}\sqrt{C^2+AG}}{2\sqrt{AB}}\right) d\varphi$$

$$m = \left(\sqrt{\frac{1}{2}}\sqrt{\frac{B}{A}}e^k\right) du + \left(i\sqrt{\frac{1}{2}}\sqrt{\frac{B}{A}}e^k\right) dv$$

$$\overline{m} = \left(\sqrt{\frac{1}{2}}\sqrt{\frac{B}{A}}e^k\right) du + \left(-i\sqrt{\frac{1}{2}}\sqrt{\frac{B}{A}}e^k\right) dv$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}\sqrt{B}C^2}{2\left(C^2+AG\right)\sqrt{A}} - \frac{\sqrt{2}\sqrt{A}\sqrt{B}G}{2\left(C^2+AG\right)} + \frac{\sqrt{2}\sqrt{B}C}{2\sqrt{C^2+AG}\sqrt{A}}\right) dt + \left(\frac{\sqrt{2}\sqrt{A}\sqrt{B}}{2\sqrt{C^2+AG}}\right) d\varphi$$

$$n = \left(-\frac{\sqrt{2}\sqrt{B}C^2}{2\left(C^2+AG\right)\sqrt{A}} - \frac{\sqrt{2}\sqrt{A}\sqrt{B}G}{2\left(C^2+AG\right)} - \frac{\sqrt{2}\sqrt{B}C}{2\sqrt{C^2+AG}\sqrt{A}}\right) dt + \left(-\frac{\sqrt{2}\sqrt{A}\sqrt{B}}{2\sqrt{C^2+AG}}\right) d\varphi$$

$$m = \left(\frac{\sqrt{2}\sqrt{A}e^{(-k)}}{2\sqrt{B}}\right) du + \left(\frac{i\sqrt{2}\sqrt{A}e^{(-k)}}{2\sqrt{B}}\right) dv$$

$$\overline{m} = \left(\frac{\sqrt{2}\sqrt{A}e^{(-k)}}{2\sqrt{B}}\right) du + \left(-\frac{i\sqrt{2}\sqrt{A}e^{(-k)}}{2\sqrt{B}}\right) dv$$

Directional Derivatives

$$D = -\frac{\sqrt{2}\sqrt{B}C^2}{2\left(C^2+AG\right)\sqrt{A}} - \frac{\sqrt{2}\sqrt{A}\sqrt{B}G}{2\left(C^2+AG\right)} + \frac{\sqrt{2}\sqrt{B}C}{2\sqrt{C^2+AG}\sqrt{A}}\,\partial_t + \frac{\sqrt{2}\sqrt{A}\sqrt{B}}{2\sqrt{C^2+AG}}\,\partial_\varphi$$

$$\Delta = -\frac{\sqrt{2}\sqrt{B}C^2}{2\left(C^2+AG\right)\sqrt{A}} - \frac{\sqrt{2}\sqrt{A}\sqrt{B}G}{2\left(C^2+AG\right)} - \frac{\sqrt{2}\sqrt{B}C}{2\sqrt{C^2+AG}\sqrt{A}}\,\partial_t - \frac{\sqrt{2}\sqrt{A}\sqrt{B}}{2\sqrt{C^2+AG}}\,\partial_\varphi$$

$$\delta = \frac{\sqrt{2}\sqrt{A}e^{(-k)}}{2\sqrt{B}}\,\partial_u + \frac{i\sqrt{2}\sqrt{A}e^{(-k)}}{2\sqrt{B}}\,\partial_v$$

$$\overline{\delta} = \frac{\sqrt{2}\sqrt{A}e^{(-k)}}{2\sqrt{B}}\,\partial_u - \frac{i\sqrt{2}\sqrt{A}e^{(-k)}}{2\sqrt{B}}\,\partial_v$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\begin{aligned}\kappa = &-\frac{\sqrt{2}C^2\frac{\partial}{\partial u}A}{4\left(C^2e^k+AGe^k\right)\sqrt{A}\sqrt{B}} - \frac{\sqrt{2}\sqrt{A}G\frac{\partial}{\partial u}A}{8\left(C^2e^k+AGe^k\right)\sqrt{B}} - \frac{i\sqrt{2}C^2\frac{\partial}{\partial v}A}{4\left(C^2e^k+AGe^k\right)\sqrt{A}\sqrt{B}} - \frac{i\sqrt{2}\sqrt{A}G\frac{\partial}{\partial v}A}{8\left(C^2e^k+AGe^k\right)\sqrt{B}} + \frac{\sqrt{2}\sqrt{A}C\frac{\partial}{\partial u}C}{4\left(C^2e^k+AGe^k\right)\sqrt{B}} + \frac{i\sqrt{2}\sqrt{A}C\frac{\partial}{\partial v}C}{4\left(C^2e^k+AGe^k\right)\sqrt{B}}\\
&+\frac{\sqrt{2}A^{\frac{3}{2}}\frac{\partial}{\partial u}G}{8\left(C^2e^k+AGe^k\right)\sqrt{B}} + \frac{i\sqrt{2}A^{\frac{3}{2}}\frac{\partial}{\partial v}G}{8\left(C^2e^k+AGe^k\right)\sqrt{B}} + \frac{\sqrt{2}\sqrt{C^2+AG}C\frac{\partial}{\partial u}A}{4\left(C^2e^k+AGe^k\right)\sqrt{A}\sqrt{B}} + \frac{i\sqrt{2}\sqrt{C^2+AG}C\frac{\partial}{\partial v}A}{4\left(C^2e^k+AGe^k\right)\sqrt{A}\sqrt{B}} - \frac{\sqrt{2}\sqrt{C^2+AG}\sqrt{A}\frac{\partial}{\partial u}C}{4\left(C^2e^k+AGe^k\right)\sqrt{B}} - \frac{i\sqrt{2}\sqrt{C^2+AG}\sqrt{A}\frac{\partial}{\partial v}C}{4\left(C^2e^k+AGe^k\right)\sqrt{B}}\end{aligned}$$

$$\mu = 0$$

$$\lambda = 0$$

$$\begin{aligned}\nu = &\frac{\sqrt{2}C^2\frac{\partial}{\partial u}A}{4\left(C^2e^k+AGe^k\right)\sqrt{A}\sqrt{B}} + \frac{\sqrt{2}\sqrt{A}G\frac{\partial}{\partial u}A}{8\left(C^2e^k+AGe^k\right)\sqrt{B}} - \frac{i\sqrt{2}C^2\frac{\partial}{\partial v}A}{4\left(C^2e^k+AGe^k\right)\sqrt{A}\sqrt{B}} - \frac{i\sqrt{2}\sqrt{A}G\frac{\partial}{\partial v}A}{8\left(C^2e^k+AGe^k\right)\sqrt{B}} - \frac{\sqrt{2}\sqrt{A}C\frac{\partial}{\partial u}C}{4\left(C^2e^k+AGe^k\right)\sqrt{B}} + \frac{i\sqrt{2}\sqrt{A}C\frac{\partial}{\partial v}C}{4\left(C^2e^k+AGe^k\right)\sqrt{B}}\\
&-\frac{\sqrt{2}A^{\frac{3}{2}}\frac{\partial}{\partial u}G}{8\left(C^2e^k+AGe^k\right)\sqrt{B}} + \frac{i\sqrt{2}A^{\frac{3}{2}}\frac{\partial}{\partial v}G}{8\left(C^2e^k+AGe^k\right)\sqrt{B}} + \frac{\sqrt{2}\sqrt{C^2+AG}C\frac{\partial}{\partial u}A}{4\left(C^2e^k+AGe^k\right)\sqrt{A}\sqrt{B}} - \frac{i\sqrt{2}\sqrt{C^2+AG}C\frac{\partial}{\partial v}A}{4\left(C^2e^k+AGe^k\right)\sqrt{A}\sqrt{B}} - \frac{\sqrt{2}\sqrt{C^2+AG}\sqrt{A}\frac{\partial}{\partial u}C}{4\left(C^2e^k+AGe^k\right)\sqrt{B}} + \frac{i\sqrt{2}\sqrt{C^2+AG}\sqrt{A}\frac{\partial}{\partial v}C}{4\left(C^2e^k+AGe^k\right)\sqrt{B}}\end{aligned}$$

$$\begin{aligned}\alpha = &-\frac{\sqrt{2}\sqrt{A}\sqrt{B}C^2\frac{\partial}{\partial u}k}{4\left(BC^2e^k+ABGe^k\right)} - \frac{\sqrt{2}A^{\frac{3}{2}}\sqrt{B}G\frac{\partial}{\partial u}k}{4\left(BC^2e^k+ABGe^k\right)} + \frac{i\sqrt{2}\sqrt{A}\sqrt{B}C^2\frac{\partial}{\partial v}k}{4\left(BC^2e^k+ABGe^k\right)} + \frac{i\sqrt{2}A^{\frac{3}{2}}\sqrt{B}G\frac{\partial}{\partial v}k}{4\left(BC^2e^k+ABGe^k\right)} + \frac{\sqrt{2}\sqrt{B}C^2\frac{\partial}{\partial u}A}{8\left(BC^2e^k+ABGe^k\right)\sqrt{A}} + \frac{\sqrt{2}\sqrt{A}\sqrt{B}G\frac{\partial}{\partial u}A}{8\left(BC^2e^k+ABGe^k\right)}\\
&-\frac{i\sqrt{2}\sqrt{B}C^2\frac{\partial}{\partial v}A}{8\left(BC^2e^k+ABGe^k\right)\sqrt{A}} - \frac{i\sqrt{2}\sqrt{A}\sqrt{B}G\frac{\partial}{\partial v}A}{8\left(BC^2e^k+ABGe^k\right)} - \frac{\sqrt{2}\sqrt{A}C^2\frac{\partial}{\partial u}B}{8\left(BC^2e^k+ABGe^k\right)\sqrt{B}} - \frac{\sqrt{2}A^{\frac{3}{2}}G\frac{\partial}{\partial u}B}{8\left(BC^2e^k+ABGe^k\right)\sqrt{B}} + \frac{i\sqrt{2}\sqrt{A}C^2\frac{\partial}{\partial v}B}{8\left(BC^2e^k+ABGe^k\right)\sqrt{B}}\\
&+\frac{i\sqrt{2}A^{\frac{3}{2}}G\frac{\partial}{\partial v}B}{8\left(BC^2e^k+ABGe^k\right)\sqrt{B}} + \frac{\sqrt{2}\sqrt{C^2+AG}\sqrt{B}C\frac{\partial}{\partial u}A}{8\left(BC^2e^k+ABGe^k\right)\sqrt{A}} - \frac{i\sqrt{2}\sqrt{C^2+AG}\sqrt{B}C\frac{\partial}{\partial v}A}{8\left(BC^2e^k+ABGe^k\right)\sqrt{A}} - \frac{\sqrt{2}\sqrt{C^2+AG}\sqrt{A}\sqrt{B}\frac{\partial}{\partial u}C}{8\left(BC^2e^k+ABGe^k\right)} + \frac{i\sqrt{2}\sqrt{C^2+AG}\sqrt{A}\sqrt{B}\frac{\partial}{\partial v}C}{8\left(BC^2e^k+ABGe^k\right)}\end{aligned}$$

$$\beta = \frac{\sqrt{2}\sqrt{A}\sqrt{B}C^2\frac{\partial}{\partial u}k}{4\left(BC^2e^k + ABGe^k\right)} + \frac{\sqrt{2}A^{\frac{3}{2}}\sqrt{B}G\frac{\partial}{\partial u}k}{4\left(BC^2e^k + ABGe^k\right)} + \frac{i\sqrt{2}\sqrt{A}\sqrt{B}C^2\frac{\partial}{\partial v}k}{4\left(BC^2e^k + ABGe^k\right)} + \frac{i\sqrt{2}A^{\frac{3}{2}}\sqrt{B}G\frac{\partial}{\partial v}k}{4\left(BC^2e^k + ABGe^k\right)} - \frac{\sqrt{2}\sqrt{B}C^2\frac{\partial}{\partial u}A}{8\left(BC^2e^k + ABGe^k\right)\sqrt{A}} - \frac{\sqrt{2}\sqrt{A}\sqrt{B}G\frac{\partial}{\partial u}A}{8\left(BC^2e^k + ABGe^k\right)}$$
$$- \frac{i\sqrt{2}\sqrt{B}C^2\frac{\partial}{\partial v}A}{8\left(BC^2e^k + ABGe^k\right)\sqrt{A}} - \frac{i\sqrt{2}\sqrt{A}\sqrt{B}G\frac{\partial}{\partial v}A}{8\left(BC^2e^k + ABGe^k\right)} + \frac{\sqrt{2}\sqrt{A}C^2\frac{\partial}{\partial u}B}{8\left(BC^2e^k + ABGe^k\right)\sqrt{B}} + \frac{\sqrt{2}A^{\frac{3}{2}}G\frac{\partial}{\partial u}B}{8\left(BC^2e^k + ABGe^k\right)\sqrt{B}} + \frac{i\sqrt{2}\sqrt{A}C^2\frac{\partial}{\partial v}B}{8\left(BC^2e^k + ABGe^k\right)\sqrt{B}}$$
$$+ \frac{i\sqrt{2}A^{\frac{3}{2}}G\frac{\partial}{\partial v}B}{8\left(BC^2e^k + ABGe^k\right)\sqrt{B}} + \frac{\sqrt{2}\sqrt{C^2 + AG}\sqrt{B}C\frac{\partial}{\partial u}A}{8\left(BC^2e^k + ABGe^k\right)\sqrt{A}} + \frac{i\sqrt{2}\sqrt{C^2 + AG}\sqrt{B}C\frac{\partial}{\partial v}A}{8\left(BC^2e^k + ABGe^k\right)\sqrt{A}} - \frac{\sqrt{2}\sqrt{C^2 + AG}\sqrt{A}\sqrt{B}\frac{\partial}{\partial u}C}{8\left(BC^2e^k + ABGe^k\right)} - \frac{i\sqrt{2}\sqrt{C^2 + AG}\sqrt{A}\sqrt{B}\frac{\partial}{\partial v}C}{8\left(BC^2e^k + ABGe^k\right)}$$

$$\tau = -\frac{\sqrt{2}\sqrt{A}\sqrt{B}G\frac{\partial}{\partial u}A}{8\left(BC^2e^k + ABGe^k\right)} - \frac{i\sqrt{2}\sqrt{A}\sqrt{B}G\frac{\partial}{\partial v}A}{8\left(BC^2e^k + ABGe^k\right)} + \frac{\sqrt{2}\sqrt{A}C^2\frac{\partial}{\partial u}B}{4\left(BC^2e^k + ABGe^k\right)\sqrt{B}} + \frac{\sqrt{2}A^{\frac{3}{2}}G\frac{\partial}{\partial u}B}{4\left(BC^2e^k + ABGe^k\right)\sqrt{B}} + \frac{i\sqrt{2}\sqrt{A}C^2\frac{\partial}{\partial v}B}{4\left(BC^2e^k + ABGe^k\right)\sqrt{B}}$$
$$+ \frac{i\sqrt{2}A^{\frac{3}{2}}G\frac{\partial}{\partial v}B}{4\left(BC^2e^k + ABGe^k\right)\sqrt{B}} - \frac{\sqrt{2}\sqrt{A}\sqrt{B}C\frac{\partial}{\partial u}C}{4\left(BC^2e^k + ABGe^k\right)} - \frac{i\sqrt{2}\sqrt{A}\sqrt{B}C\frac{\partial}{\partial v}C}{4\left(BC^2e^k + ABGe^k\right)} - \frac{\sqrt{2}A^{\frac{3}{2}}\sqrt{B}\frac{\partial}{\partial u}G}{8\left(BC^2e^k + ABGe^k\right)} - \frac{i\sqrt{2}A^{\frac{3}{2}}\sqrt{B}\frac{\partial}{\partial v}G}{8\left(BC^2e^k + ABGe^k\right)}$$

$$\pi = \frac{\sqrt{2}\sqrt{A}\sqrt{B}G\frac{\partial}{\partial u}A}{8\left(BC^2e^k + ABGe^k\right)} - \frac{i\sqrt{2}\sqrt{A}\sqrt{B}G\frac{\partial}{\partial v}A}{8\left(BC^2e^k + ABGe^k\right)} - \frac{\sqrt{2}\sqrt{A}C^2\frac{\partial}{\partial u}B}{4\left(BC^2e^k + ABGe^k\right)\sqrt{B}} - \frac{\sqrt{2}A^{\frac{3}{2}}G\frac{\partial}{\partial u}B}{4\left(BC^2e^k + ABGe^k\right)\sqrt{B}} + \frac{i\sqrt{2}\sqrt{A}C^2\frac{\partial}{\partial v}B}{4\left(BC^2e^k + ABGe^k\right)\sqrt{B}}$$
$$+ \frac{i\sqrt{2}A^{\frac{3}{2}}G\frac{\partial}{\partial v}B}{4\left(BC^2e^k + ABGe^k\right)\sqrt{B}} + \frac{\sqrt{2}\sqrt{A}\sqrt{B}C\frac{\partial}{\partial u}C}{4\left(BC^2e^k + ABGe^k\right)} - \frac{i\sqrt{2}\sqrt{A}\sqrt{B}C\frac{\partial}{\partial v}C}{4\left(BC^2e^k + ABGe^k\right)} + \frac{\sqrt{2}A^{\frac{3}{2}}\sqrt{B}\frac{\partial}{\partial u}G}{8\left(BC^2e^k + ABGe^k\right)} - \frac{i\sqrt{2}A^{\frac{3}{2}}\sqrt{B}\frac{\partial}{\partial v}G}{8\left(BC^2e^k + ABGe^k\right)}$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$
\begin{aligned}
\Phi_{00} = & -\frac{BC^2G\frac{\partial}{\partial u}A^2}{8\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} - \frac{ABG^2\frac{\partial}{\partial u}A^2}{16\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} + \frac{BC^4\frac{\partial^2}{(\partial u)^2}A}{4\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} \\
& + \frac{3\,ABC^2G\frac{\partial^2}{(\partial u)^2}A}{8\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} + \frac{A^2BG^2\frac{\partial^2}{(\partial u)^2}A}{8\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} - \frac{BC^2G\frac{\partial}{\partial v}A^2}{8\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} \\
& - \frac{ABG^2\frac{\partial}{\partial v}A^2}{16\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} + \frac{BC^4\frac{\partial^2}{(\partial v)^2}A}{4\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} + \frac{3\,ABC^2G\frac{\partial^2}{(\partial v)^2}A}{8\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} \\
& + \frac{A^2BG^2\frac{\partial^2}{(\partial v)^2}A}{8\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} - \frac{C^4\frac{\partial}{\partial u}A\frac{\partial}{\partial u}B}{4\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} - \frac{3\,AC^2G\frac{\partial}{\partial u}A\frac{\partial}{\partial u}B}{8\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} \\
& - \frac{A^2G^2\frac{\partial}{\partial u}A\frac{\partial}{\partial u}B}{8\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} - \frac{C^4\frac{\partial}{\partial v}A\frac{\partial}{\partial v}B}{4\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} - \frac{3\,AC^2G\frac{\partial}{\partial v}A\frac{\partial}{\partial v}B}{8\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} \\
& - \frac{A^2G^2\frac{\partial}{\partial v}A\frac{\partial}{\partial v}B}{8\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} - \frac{BC^3\frac{\partial}{\partial u}A\frac{\partial}{\partial u}C}{4\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} + \frac{AC^3\frac{\partial}{\partial u}B\frac{\partial}{\partial u}C}{4\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} \\
& + \frac{A^2CG\frac{\partial}{\partial u}B\frac{\partial}{\partial u}C}{4\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} + \frac{ABC^2\frac{\partial}{\partial u}C^2}{4\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} - \frac{ABC^3\frac{\partial^2}{(\partial u)^2}C}{4\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} \\
& - \frac{A^2BCG\frac{\partial^2}{(\partial u)^2}C}{4\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} - \frac{BC^3\frac{\partial}{\partial v}A\frac{\partial}{\partial v}C}{4\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} + \dots
\end{aligned}
$$

$$\Phi_{01} = 0$$

$$
\begin{aligned}
\Phi_{02} = & \frac{AB^2C^2G\frac{\partial}{\partial u}A\frac{\partial}{\partial u}k}{4\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} + \frac{A^2B^2G^2\frac{\partial}{\partial u}A\frac{\partial}{\partial u}k}{4\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} + \frac{i\,AB^2C^2G\frac{\partial}{\partial v}A\frac{\partial}{\partial u}k}{4\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} \\
& + \frac{i\,A^2B^2G^2\frac{\partial}{\partial v}A\frac{\partial}{\partial u}k}{4\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} - \frac{ABC^4\frac{\partial}{\partial u}B\frac{\partial}{\partial u}k}{2\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} - \frac{A^2BC^2G\frac{\partial}{\partial u}B\frac{\partial}{\partial u}k}{B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}} \\
& - \frac{A^3BG^2\frac{\partial}{\partial u}B\frac{\partial}{\partial u}k}{2\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} - \frac{i\,ABC^4\frac{\partial}{\partial v}B\frac{\partial}{\partial u}k}{2\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} - \frac{i\,A^2BC^2G\frac{\partial}{\partial v}B\frac{\partial}{\partial u}k}{B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}} \\
& - \frac{i\,A^3BG^2\frac{\partial}{\partial v}B\frac{\partial}{\partial u}k}{2\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} + \frac{AB^2C^3\frac{\partial}{\partial u}C\frac{\partial}{\partial u}k}{2\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} + \frac{A^2B^2CG\frac{\partial}{\partial u}C\frac{\partial}{\partial u}k}{2\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} \\
& + \frac{i\,AB^2C^3\frac{\partial}{\partial v}C\frac{\partial}{\partial u}k}{2\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} + \frac{i\,A^2B^2CG\frac{\partial}{\partial v}C\frac{\partial}{\partial u}k}{2\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} + \frac{A^2B^2C^2\frac{\partial}{\partial u}G\frac{\partial}{\partial u}k}{4\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} \\
& + \frac{A^3B^2G\frac{\partial}{\partial u}G\frac{\partial}{\partial u}k}{4\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} + \frac{i\,A^2B^2C^2\frac{\partial}{\partial v}G\frac{\partial}{\partial u}k}{4\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} + \frac{i\,A^3B^2G\frac{\partial}{\partial v}G\frac{\partial}{\partial u}k}{4\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} \\
& + \frac{i\,AB^2C^2G\frac{\partial}{\partial u}A\frac{\partial}{\partial v}k}{4\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} + \frac{i\,A^2B^2G^2\frac{\partial}{\partial u}A\frac{\partial}{\partial v}k}{4\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} - \frac{AB^2C^2G\frac{\partial}{\partial v}A\frac{\partial}{\partial v}k}{4\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} - \dots
\end{aligned}
$$

$\Phi_{10} = 0$

$$
\begin{aligned}
\Phi_{11} = & -\frac{A^2B^2C^2\frac{\partial^2}{(\partial u)^2}k}{4\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} - \frac{A^3B^2G\frac{\partial^2}{(\partial u)^2}k}{4\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} - \frac{A^2B^2C^2\frac{\partial^2}{(\partial v)^2}k}{4\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} - \frac{A^3B^2G\frac{\partial^2}{(\partial v)^2}k}{4\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} \\
& - \frac{B^2C^2\frac{\partial}{\partial u}A^2}{8\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} - \frac{AB^2G\frac{\partial}{\partial u}A^2}{8\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} + \frac{AB^2C^2\frac{\partial^2}{(\partial u)^2}A}{8\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} + \frac{A^2B^2G\frac{\partial^2}{(\partial u)^2}A}{8\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} \\
& - \frac{B^2C^2\frac{\partial}{\partial v}A^2}{8\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} - \frac{AB^2G\frac{\partial}{\partial v}A^2}{8\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} + \frac{AB^2C^2\frac{\partial^2}{(\partial v)^2}A}{8\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} + \frac{A^2B^2G\frac{\partial^2}{(\partial v)^2}A}{8\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} \\
& - \frac{A^2BG\frac{\partial}{\partial u}A\frac{\partial}{\partial u}B}{16\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} + \frac{3\,A^2C^2\frac{\partial}{\partial u}B^2}{16\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} + \frac{3\,A^3G\frac{\partial}{\partial u}B^2}{16\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} \\
& - \frac{A^2BC^2\frac{\partial^2}{(\partial u)^2}B}{8\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} - \frac{A^3BG\frac{\partial^2}{(\partial u)^2}B}{8\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} - \frac{A^2BG\frac{\partial}{\partial v}A\frac{\partial}{\partial v}B}{16\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} + \frac{3\,A^2C^2\frac{\partial}{\partial v}B^2}{16\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} \\
& + \frac{3\,A^3G\frac{\partial}{\partial v}B^2}{16\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} - \frac{A^2BC^2\frac{\partial^2}{(\partial v)^2}B}{8\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} - \frac{A^3BG\frac{\partial^2}{(\partial v)^2}B}{8\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} - \frac{A^2BC\frac{\partial}{\partial u}B\frac{\partial}{\partial u}C}{8\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} \\
& + \frac{A^2B^2\frac{\partial}{\partial u}C^2}{16\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} - \frac{A^2BC\frac{\partial}{\partial v}B\frac{\partial}{\partial v}C}{8\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} + \frac{A^2B^2\frac{\partial}{\partial v}C^2}{16\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} + \frac{A^2B^2\frac{\partial}{\partial u}A\frac{\partial}{\partial u}G}{16\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} \\
& - \frac{A^3B\frac{\partial}{\partial u}B\frac{\partial}{\partial u}G}{16\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} + \frac{A^2B^2\frac{\partial}{\partial v}A\frac{\partial}{\partial v}G}{16\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)} - \frac{A^3B\frac{\partial}{\partial v}B\frac{\partial}{\partial v}G}{16\left(AB^3C^2e^{(2\,k)}+A^2B^3Ge^{(2\,k)}\right)}
\end{aligned}
$$

$\Phi_{12} = 0$

$$
\begin{aligned}
\Phi_{20} = {} & \frac{AB^2C^2G\frac{\partial}{\partial u}A\frac{\partial}{\partial u}k}{4\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} + \frac{A^2B^2G^2\frac{\partial}{\partial u}A\frac{\partial}{\partial u}k}{4\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} - \frac{i\,AB^2C^2G\frac{\partial}{\partial v}A\frac{\partial}{\partial u}k}{4\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} \\
& - \frac{i\,A^2B^2G^2\frac{\partial}{\partial v}A\frac{\partial}{\partial u}k}{4\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} - \frac{ABC^4\frac{\partial}{\partial u}B\frac{\partial}{\partial u}k}{2\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} - \frac{A^2BC^2G\frac{\partial}{\partial u}B\frac{\partial}{\partial u}k}{B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}} \\
& - \frac{A^3BG^2\frac{\partial}{\partial u}B\frac{\partial}{\partial u}k}{2\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} + \frac{i\,ABC^4\frac{\partial}{\partial v}B\frac{\partial}{\partial u}k}{2\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} + \frac{i\,A^2BC^2G\frac{\partial}{\partial v}B\frac{\partial}{\partial u}k}{B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}} \\
& + \frac{i\,A^3BG^2\frac{\partial}{\partial v}B\frac{\partial}{\partial u}k}{2\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} + \frac{AB^2C^3\frac{\partial}{\partial u}C\frac{\partial}{\partial u}k}{2\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} + \frac{A^2B^2CG\frac{\partial}{\partial u}C\frac{\partial}{\partial u}k}{2\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} \\
& - \frac{i\,AB^2C^3\frac{\partial}{\partial v}C\frac{\partial}{\partial u}k}{2\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} - \frac{i\,A^2B^2CG\frac{\partial}{\partial v}C\frac{\partial}{\partial u}k}{2\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} + \frac{A^2B^2C^2\frac{\partial}{\partial u}G\frac{\partial}{\partial u}k}{4\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} \\
& + \frac{A^3B^2G\frac{\partial}{\partial u}G\frac{\partial}{\partial u}k}{4\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} - \frac{i\,A^2B^2C^2\frac{\partial}{\partial v}G\frac{\partial}{\partial u}k}{4\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} - \frac{i\,A^3B^2G\frac{\partial}{\partial v}G\frac{\partial}{\partial u}k}{4\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} \\
& - \frac{i\,AB^2C^2G\frac{\partial}{\partial u}A\frac{\partial}{\partial v}k}{4\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} - \frac{i\,A^2B^2G^2\frac{\partial}{\partial u}A\frac{\partial}{\partial v}k}{4\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} - \frac{AB^2C^2G\frac{\partial}{\partial v}A\frac{\partial}{\partial v}k}{4\left(B^3C^4e^{(2\,k)}+2\,AB^3C^2Ge^{(2\,k)}+A^2B^3G^2e^{(2\,k)}\right)} - \cdots
\end{aligned}
$$

$$\Phi_{21} = 0$$

$$
\begin{aligned}
\Phi_{22} = {} & -\frac{BC^2G\frac{\partial}{\partial u}A^2}{8\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} - \frac{ABG^2\frac{\partial}{\partial u}A^2}{16\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} + \frac{BC^4\frac{\partial^2}{(\partial u)^2}A}{4\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} \\
& + \frac{3\,ABC^2G\frac{\partial^2}{(\partial u)^2}A}{8\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} + \frac{A^2BG^2\frac{\partial^2}{(\partial u)^2}A}{8\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} - \frac{BC^2G\frac{\partial}{\partial v}A^2}{8\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} \\
& - \frac{ABG^2\frac{\partial}{\partial v}A^2}{16\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} + \frac{BC^4\frac{\partial^2}{(\partial v)^2}A}{4\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} + \frac{3\,ABC^2G\frac{\partial^2}{(\partial v)^2}A}{8\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} \\
& + \frac{A^2BG^2\frac{\partial^2}{(\partial v)^2}A}{8\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} - \frac{C^4\frac{\partial}{\partial u}A\frac{\partial}{\partial u}B}{4\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} - \frac{3\,AC^2G\frac{\partial}{\partial u}A\frac{\partial}{\partial u}B}{8\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} \\
& - \frac{A^2G^2\frac{\partial}{\partial u}A\frac{\partial}{\partial u}B}{8\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} - \frac{C^4\frac{\partial}{\partial v}A\frac{\partial}{\partial v}B}{4\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} - \frac{3\,AC^2G\frac{\partial}{\partial v}A\frac{\partial}{\partial v}B}{8\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} \\
& - \frac{A^2G^2\frac{\partial}{\partial v}A\frac{\partial}{\partial v}B}{8\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} - \frac{BC^3\frac{\partial}{\partial u}A\frac{\partial}{\partial u}C}{4\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} + \frac{AC^3\frac{\partial}{\partial u}B\frac{\partial}{\partial u}C}{4\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} \\
& + \frac{A^2CG\frac{\partial}{\partial u}B\frac{\partial}{\partial u}C}{4\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} + \frac{ABC^2\frac{\partial}{\partial u}C^2}{4\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} - \frac{ABC^3\frac{\partial^2}{(\partial u)^2}C}{4\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} \\
& - \frac{A^2BCG\frac{\partial^2}{(\partial u)^2}C}{4\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} - \frac{BC^3\frac{\partial}{\partial v}A\frac{\partial}{\partial v}C}{4\left(B^2C^4e^{(2\,k)}+2\,AB^2C^2Ge^{(2\,k)}+A^2B^2G^2e^{(2\,k)}\right)} + \cdots
\end{aligned}
$$

$$
\begin{aligned}
\Lambda = & -\frac{A^2 B^2 C^4 \frac{\partial^2}{(\partial u)^2} k}{12 \left(A B^3 C^4 e^{(2\,k)} + 2\,A^2 B^3 C^2 G e^{(2\,k)} + A^3 B^3 G^2 e^{(2\,k)}\right)} - \frac{A^3 B^2 C^2 G \frac{\partial^2}{(\partial u)^2} k}{6 \left(A B^3 C^4 e^{(2\,k)} + 2\,A^2 B^3 C^2 G e^{(2\,k)} + A^3 B^3 G^2 e^{(2\,k)}\right)} - \frac{A^4 B^2 G^2 \frac{\partial^2}{(\partial u)^2} k}{12 \left(A B^3 C^4 e^{(2\,k)} + 2\,A^2 B^3 C^2 G e^{(2\,k)} + A^3 B^3 G^2 e^{(2\,k)}\right)} \\
& - \frac{A^2 B^2 C^4 \frac{\partial^2}{(\partial v)^2} k}{12 \left(A B^3 C^4 e^{(2\,k)} + 2\,A^2 B^3 C^2 G e^{(2\,k)} + A^3 B^3 G^2 e^{(2\,k)}\right)} - \frac{A^3 B^2 C^2 G \frac{\partial^2}{(\partial v)^2} k}{6 \left(A B^3 C^4 e^{(2\,k)} + 2\,A^2 B^3 C^2 G e^{(2\,k)} + A^3 B^3 G^2 e^{(2\,k)}\right)} - \frac{A^4 B^2 G^2 \frac{\partial^2}{(\partial v)^2} k}{12 \left(A B^3 C^4 e^{(2\,k)} + 2\,A^2 B^3 C^2 G e^{(2\,k)} + A^3 B^3 G^2 e^{(2\,k)}\right)} \\
& - \frac{B^2 C^4 \frac{\partial}{\partial u} A^2}{24 \left(A B^3 C^4 e^{(2\,k)} + 2\,A^2 B^3 C^2 G e^{(2\,k)} + A^3 B^3 G^2 e^{(2\,k)}\right)} - \frac{A B^2 C^2 G \frac{\partial}{\partial u} A^2}{12 \left(A B^3 C^4 e^{(2\,k)} + 2\,A^2 B^3 C^2 G e^{(2\,k)} + A^3 B^3 G^2 e^{(2\,k)}\right)} - \frac{A^2 B^2 G^2 \frac{\partial}{\partial u} A^2}{48 \left(A B^3 C^4 e^{(2\,k)} + 2\,A^2 B^3 C^2 G e^{(2\,k)} + A^3 B^3 G^2 e^{(2\,k)}\right)} \\
& + \frac{A B^2 C^4 \frac{\partial^2}{(\partial u)^2} A}{24 \left(A B^3 C^4 e^{(2\,k)} + 2\,A^2 B^3 C^2 G e^{(2\,k)} + A^3 B^3 G^2 e^{(2\,k)}\right)} + \frac{A^2 B^2 C^2 G \frac{\partial^2}{(\partial u)^2} A}{24 \left(A B^3 C^4 e^{(2\,k)} + 2\,A^2 B^3 C^2 G e^{(2\,k)} + A^3 B^3 G^2 e^{(2\,k)}\right)} \\
& - \frac{B^2 C^4 \frac{\partial}{\partial v} A^2}{24 \left(A B^3 C^4 e^{(2\,k)} + 2\,A^2 B^3 C^2 G e^{(2\,k)} + A^3 B^3 G^2 e^{(2\,k)}\right)} - \frac{A B^2 C^2 G \frac{\partial}{\partial v} A^2}{12 \left(A B^3 C^4 e^{(2\,k)} + 2\,A^2 B^3 C^2 G e^{(2\,k)} + A^3 B^3 G^2 e^{(2\,k)}\right)} - \frac{A^2 B^2 G^2 \frac{\partial}{\partial v} A^2}{48 \left(A B^3 C^4 e^{(2\,k)} + 2\,A^2 B^3 C^2 G e^{(2\,k)} + A^3 B^3 G^2 e^{(2\,k)}\right)} \\
& + \frac{A B^2 C^4 \frac{\partial^2}{(\partial v)^2} A}{24 \left(A B^3 C^4 e^{(2\,k)} + 2\,A^2 B^3 C^2 G e^{(2\,k)} + A^3 B^3 G^2 e^{(2\,k)}\right)} + \frac{A^2 B^2 C^2 G \frac{\partial^2}{(\partial v)^2} A}{24 \left(A B^3 C^4 e^{(2\,k)} + 2\,A^2 B^3 C^2 G e^{(2\,k)} + A^3 B^3 G^2 e^{(2\,k)}\right)} + \frac{A^2 B C^2 G \frac{\partial}{\partial u} A \frac{\partial}{\partial u} B}{16 \left(A B^3 C^4 e^{(2\,k)} + 2\,A^2 B^3 C^2 G e^{(2\,k)} + A^3 B^3 G^2 e^{(2\,k)}\right)} \\
& + \frac{A^3 B G^2 \frac{\partial}{\partial u} A \frac{\partial}{\partial u} B}{16 \left(A B^3 C^4 e^{(2\,k)} + 2\,A^2 B^3 C^2 G e^{(2\,k)} + A^3 B^3 G^2 e^{(2\,k)}\right)} - \frac{5\,A^2 C^4 \frac{\partial}{\partial u} B^2}{48 \left(A B^3 C^4 e^{(2\,k)} + 2\,A^2 B^3 C^2 G e^{(2\,k)} + A^3 B^3 G^2 e^{(2\,k)}\right)} - \frac{5\,A^3 C^2 G \frac{\partial}{\partial u} B^2}{24 \left(A B^3 C^4 e^{(2\,k)} + 2\,A^2 B^3 C^2 G e^{(2\,k)} + A^3 B^3 G^2 e^{(2\,k)}\right)} \\
& - \frac{5\,A^4 G^2 \frac{\partial}{\partial u} B^2}{48 \left(A B^3 C^4 e^{(2\,k)} + 2\,A^2 B^3 C^2 G e^{(2\,k)} + A^3 B^3 G^2 e^{(2\,k)}\right)} + \frac{A^2 B C^4 \frac{\partial^2}{(\partial u)^2} B}{24 \left(A B^3 C^4 e^{(2\,k)} + 2\,A^2 B^3 C^2 G e^{(2\,k)} + A^3 B^3 G^2 e^{(2\,k)}\right)} + \ldots
\end{aligned}
$$

Weyl Tensor Components

$$
\begin{aligned}
\Psi_0 = & -\frac{ABC^4 \frac{\partial}{\partial u} A \frac{\partial}{\partial u} k}{2\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} - \frac{3\,A^2BC^2G \frac{\partial}{\partial u} A \frac{\partial}{\partial u} k}{4\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} - \frac{A^3BG^2 \frac{\partial}{\partial u} A \frac{\partial}{\partial u} k}{4\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} \\
& - \frac{i\,ABC^4 \frac{\partial}{\partial v} A \frac{\partial}{\partial u} k}{2\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} - \frac{3i\,A^2BC^2G \frac{\partial}{\partial v} A \frac{\partial}{\partial u} k}{4\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} - \frac{i\,A^3BG^2 \frac{\partial}{\partial v} A \frac{\partial}{\partial u} k}{4\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} \\
& + \frac{A^2BC^3 \frac{\partial}{\partial u} C \frac{\partial}{\partial u} k}{2\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} + \frac{A^3BCG \frac{\partial}{\partial u} C \frac{\partial}{\partial u} k}{2\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} \\
& + \frac{i\,A^2BC^3 \frac{\partial}{\partial v} C \frac{\partial}{\partial u} k}{2\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} + \frac{i\,A^3BCG \frac{\partial}{\partial v} C \frac{\partial}{\partial u} k}{2\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} \\
& + \frac{A^3BC^2 \frac{\partial}{\partial u} G \frac{\partial}{\partial u} k}{4\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} + \frac{A^4BG \frac{\partial}{\partial u} G \frac{\partial}{\partial u} k}{4\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} + \frac{i\,A^3BC^2 \frac{\partial}{\partial v} G \frac{\partial}{\partial u} k}{4\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} \\
& + \frac{i\,A^4BG \frac{\partial}{\partial v} G \frac{\partial}{\partial u} k}{4\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} - \frac{i\,ABC^4 \frac{\partial}{\partial u} A \frac{\partial}{\partial v} k}{2\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} - \frac{3i\,A^2BC^2G \frac{\partial}{\partial u} A \frac{\partial}{\partial v} k}{4\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} \\
& - \frac{i\,A^3BG^2 \frac{\partial}{\partial u} A \frac{\partial}{\partial v} k}{4\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} + \frac{ABC^4 \frac{\partial}{\partial v} A \frac{\partial}{\partial v} k}{2\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} + \frac{3\,A^2BC^2G \frac{\partial}{\partial v} A \frac{\partial}{\partial v} k}{4\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} \\
& + \frac{A^3BG^2 \frac{\partial}{\partial v} A \frac{\partial}{\partial v} k}{4\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} + \frac{i\,A^2BC^3 \frac{\partial}{\partial u} C \frac{\partial}{\partial v} k}{2\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} + \ldots
\end{aligned}
$$

$$\Psi_1 = 0$$

$$
\begin{aligned}
\Psi_2 = {} & \frac{A^2B^2C^4\frac{\partial^2}{(\partial u)^2}k}{6\left(AB^3C^4e^{(2\,k)}+2\,A^2B^3C^2Ge^{(2\,k)}+A^3B^3G^2e^{(2\,k)}\right)} + \frac{A^3B^2C^2G\frac{\partial^2}{(\partial u)^2}k}{3\left(AB^3C^4e^{(2\,k)}+2\,A^2B^3C^2Ge^{(2\,k)}+A^3B^3G^2e^{(2\,k)}\right)} + \frac{A^4B^2G^2\frac{\partial^2}{(\partial u)^2}k}{6\left(AB^3C^4e^{(2\,k)}+2\,A^2B^3C^2Ge^{(2\,k)}+A^3B^3G^2e^{(2\,k)}\right)} \\
& + \frac{A^2B^2C^4\frac{\partial^2}{(\partial v)^2}k}{6\left(AB^3C^4e^{(2\,k)}+2\,A^2B^3C^2Ge^{(2\,k)}+A^3B^3G^2e^{(2\,k)}\right)} + \frac{A^3B^2C^2G\frac{\partial^2}{(\partial v)^2}k}{3\left(AB^3C^4e^{(2\,k)}+2\,A^2B^3C^2Ge^{(2\,k)}+A^3B^3G^2e^{(2\,k)}\right)} + \frac{A^4B^2G^2\frac{\partial^2}{(\partial v)^2}k}{6\left(AB^3C^4e^{(2\,k)}+2\,A^2B^3C^2Ge^{(2\,k)}+A^3B^3G^2e^{(2\,k)}\right)} \\
& + \frac{B^2C^4\frac{\partial}{\partial u}A^2}{12\left(AB^3C^4e^{(2\,k)}+2\,A^2B^3C^2Ge^{(2\,k)}+A^3B^3G^2e^{(2\,k)}\right)} + \frac{AB^2C^2G\frac{\partial}{\partial u}A^2}{6\left(AB^3C^4e^{(2\,k)}+2\,A^2B^3C^2Ge^{(2\,k)}+A^3B^3G^2e^{(2\,k)}\right)} + \frac{5\,A^2B^2G^2\frac{\partial}{\partial u}A^2}{48\left(AB^3C^4e^{(2\,k)}+2\,A^2B^3C^2Ge^{(2\,k)}+A^3B^3G^2e^{(2\,k)}\right)} \\
& - \frac{AB^2C^4\frac{\partial^2}{(\partial u)^2}A}{12\left(AB^3C^4e^{(2\,k)}+2\,A^2B^3C^2Ge^{(2\,k)}+A^3B^3G^2e^{(2\,k)}\right)} - \frac{5\,A^2B^2C^2G\frac{\partial^2}{(\partial u)^2}A}{24\left(AB^3C^4e^{(2\,k)}+2\,A^2B^3C^2Ge^{(2\,k)}+A^3B^3G^2e^{(2\,k)}\right)} - \frac{A^3B^2G^2\frac{\partial^2}{(\partial u)^2}A}{8\left(AB^3C^4e^{(2\,k)}+2\,A^2B^3C^2Ge^{(2\,k)}+A^3B^3G^2e^{(2\,k)}\right)} \\
& + \frac{B^2C^4\frac{\partial}{\partial v}A^2}{12\left(AB^3C^4e^{(2\,k)}+2\,A^2B^3C^2Ge^{(2\,k)}+A^3B^3G^2e^{(2\,k)}\right)} + \frac{AB^2C^2G\frac{\partial}{\partial v}A^2}{6\left(AB^3C^4e^{(2\,k)}+2\,A^2B^3C^2Ge^{(2\,k)}+A^3B^3G^2e^{(2\,k)}\right)} + \frac{5\,A^2B^2G^2\frac{\partial}{\partial v}A^2}{48\left(AB^3C^4e^{(2\,k)}+2\,A^2B^3C^2Ge^{(2\,k)}+A^3B^3G^2e^{(2\,k)}\right)} \\
& - \frac{AB^2C^4\frac{\partial^2}{(\partial v)^2}A}{12\left(AB^3C^4e^{(2\,k)}+2\,A^2B^3C^2Ge^{(2\,k)}+A^3B^3G^2e^{(2\,k)}\right)} - \frac{5\,A^2B^2C^2G\frac{\partial^2}{(\partial v)^2}A}{24\left(AB^3C^4e^{(2\,k)}+2\,A^2B^3C^2Ge^{(2\,k)}+A^3B^3G^2e^{(2\,k)}\right)} - \frac{A^3B^2G^2\frac{\partial^2}{(\partial v)^2}A}{8\left(AB^3C^4e^{(2\,k)}+2\,A^2B^3C^2Ge^{(2\,k)}+A^3B^3G^2e^{(2\,k)}\right)} \\
& - \frac{A^2C^4\frac{\partial}{\partial u}B^2}{6\left(AB^3C^4e^{(2\,k)}+2\,A^2B^3C^2Ge^{(2\,k)}+A^3B^3G^2e^{(2\,k)}\right)} - \frac{A^3C^2G\frac{\partial}{\partial u}B^2}{3\left(AB^3C^4e^{(2\,k)}+2\,A^2B^3C^2Ge^{(2\,k)}+A^3B^3G^2e^{(2\,k)}\right)} - \frac{A^4G^2\frac{\partial}{\partial u}B^2}{6\left(AB^3C^4e^{(2\,k)}+2\,A^2B^3C^2Ge^{(2\,k)}+A^3B^3G^2e^{(2\,k)}\right)} \\
& + \frac{A^2BC^4\frac{\partial^2}{(\partial u)^2}B}{6\left(AB^3C^4e^{(2\,k)}+2\,A^2B^3C^2Ge^{(2\,k)}+A^3B^3G^2e^{(2\,k)}\right)} + \frac{A^3BC^2G\frac{\partial^2}{(\partial u)^2}B}{3\left(AB^3C^4e^{(2\,k)}+2\,A^2B^3C^2Ge^{(2\,k)}+A^3B^3G^2e^{(2\,k)}\right)} + \ldots
\end{aligned}
$$

$$\Psi_3 = 0$$

$$
\begin{aligned}
\Psi_4 = & -\frac{ABC^4 \frac{\partial}{\partial u} A \frac{\partial}{\partial u} k}{2\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} - \frac{3\,A^2BC^2G \frac{\partial}{\partial u} A \frac{\partial}{\partial u} k}{4\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} - \frac{A^3BG^2 \frac{\partial}{\partial u} A \frac{\partial}{\partial u} k}{4\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} \\
& + \frac{i\,ABC^4 \frac{\partial}{\partial v} A \frac{\partial}{\partial u} k}{2\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} + \frac{3i\,A^2BC^2G \frac{\partial}{\partial v} A \frac{\partial}{\partial u} k}{4\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} + \frac{i\,A^3BG^2 \frac{\partial}{\partial v} A \frac{\partial}{\partial u} k}{4\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} \\
& + \frac{A^2BC^3 \frac{\partial}{\partial u} C \frac{\partial}{\partial u} k}{2\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} + \frac{A^3BCG \frac{\partial}{\partial u} C \frac{\partial}{\partial u} k}{2\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} - \frac{i\,A^2BC^3 \frac{\partial}{\partial v} C \frac{\partial}{\partial u} k}{2\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} \\
& - \frac{i\,A^3BCG \frac{\partial}{\partial v} C \frac{\partial}{\partial u} k}{2\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} + \frac{A^3BC^2 \frac{\partial}{\partial u} G \frac{\partial}{\partial u} k}{4\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} + \frac{A^4BG \frac{\partial}{\partial u} G \frac{\partial}{\partial u} k}{4\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} \\
& - \frac{i\,A^3BC^2 \frac{\partial}{\partial v} G \frac{\partial}{\partial u} k}{4\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} - \frac{i\,A^4BG \frac{\partial}{\partial v} G \frac{\partial}{\partial u} k}{4\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} + \frac{i\,ABC^4 \frac{\partial}{\partial u} A \frac{\partial}{\partial v} k}{2\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} \\
& + \frac{3i\,A^2BC^2G \frac{\partial}{\partial u} A \frac{\partial}{\partial v} k}{4\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} + \frac{i\,A^3BG^2 \frac{\partial}{\partial u} A \frac{\partial}{\partial v} k}{4\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} \\
& + \frac{ABC^4 \frac{\partial}{\partial v} A \frac{\partial}{\partial v} k}{2\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} + \frac{3\,A^2BC^2G \frac{\partial}{\partial v} A \frac{\partial}{\partial v} k}{4\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} \\
& + \frac{A^3BG^2 \frac{\partial}{\partial v} A \frac{\partial}{\partial v} k}{4\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} - \frac{i\,A^2BC^3 \frac{\partial}{\partial u} C \frac{\partial}{\partial v} k}{2\left(AB^2C^4e^{(2\,k)} + 2\,A^2B^2C^2Ge^{(2\,k)} + A^3B^2G^2e^{(2\,k)}\right)} - \cdots
\end{aligned}
$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"